Visual Anagrams: Generating Multi-View Optical Illusions with Diffusion Models

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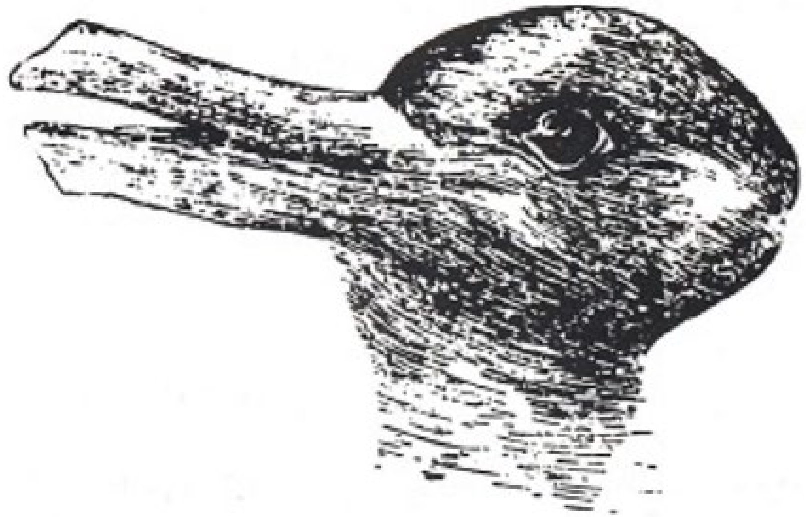
Introduction: Optical Illusions

What we see is not necessarily true, brain can perceive different pattern from the same signal.



Duck or Rabbit

Jesus or Girls

Campfire or Man

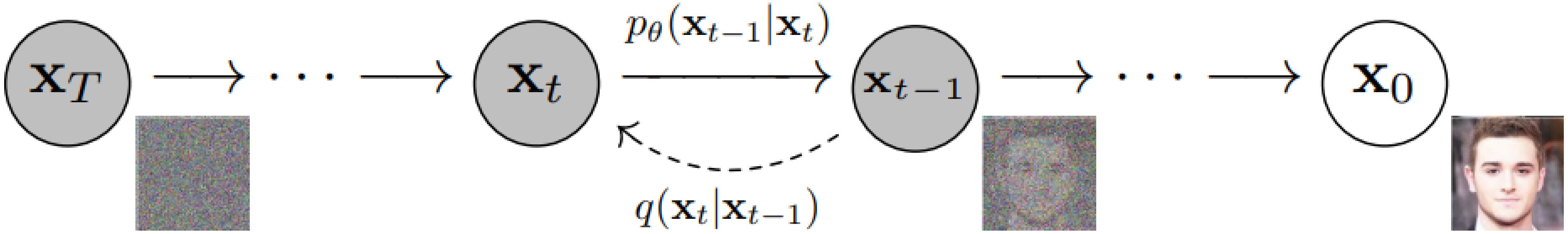
Introduction: Diffusion

Denoising Diffusion Probabilistic Models (*Ben Mildenhall et al.*, 2020)

The foundational work that treat denoising process as a Markov chain and predict noise at each step.

Denoising Diffusion Implicit Models (*Jiaming Song et al*., 2020)

treats diffusion as a continuous process governed by Stochastic Differential Equations, then the model learn the gradient of the data distribution.



$x_T \sim \mathcal{N}(0, I)$

$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$

Introduction: Motivation

Can we generate optical illusions with diffusion models in a zero-shot manner?

Yes!

Visual Anagrams / Factorized Diffusion

photo of a volcano

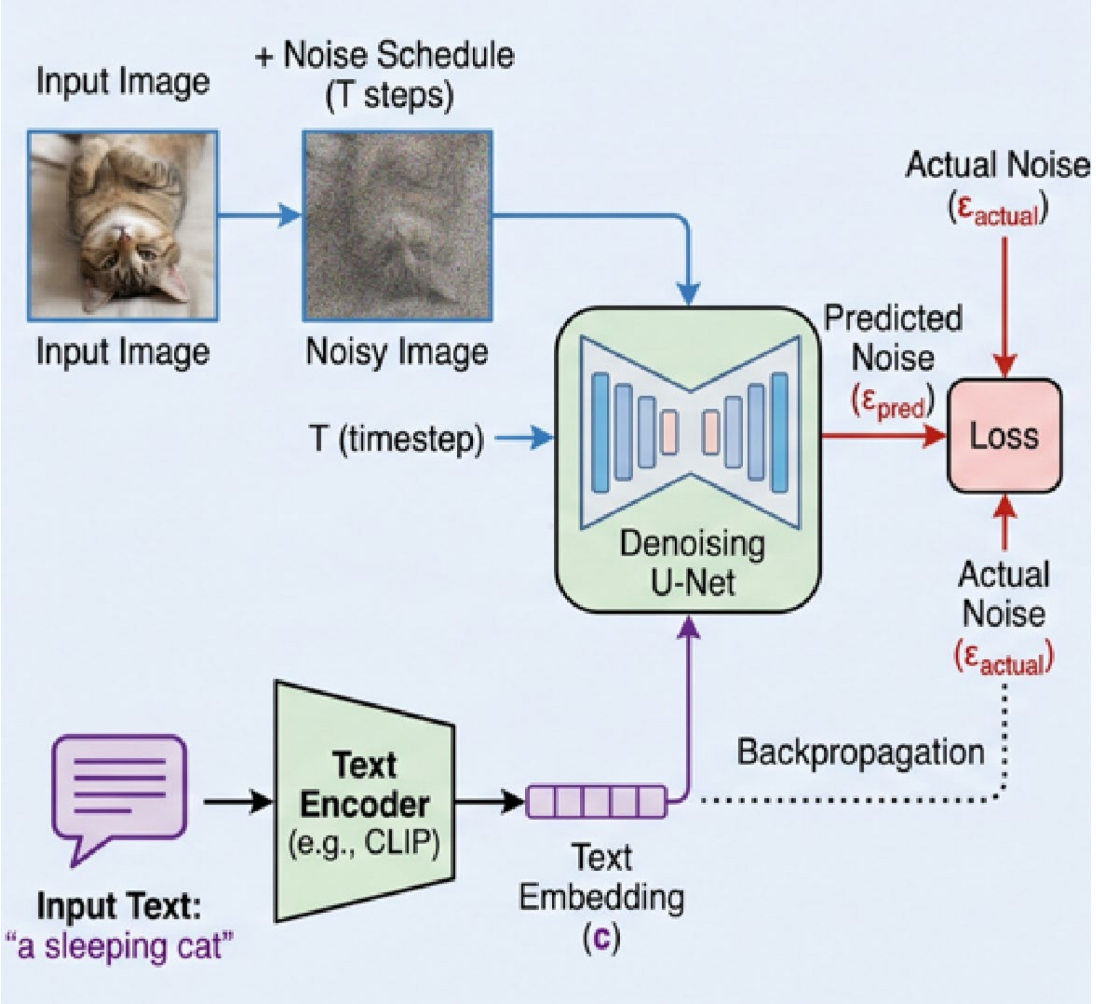
photo of a face

photo of a man

photo of a woman

Methods

Text-conditioned Diffusion Model



$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]$$

$$\epsilon_\theta(x_t, t)$$

Unconditioned

$$\epsilon_\theta(x_t, t, c)$$

Text-conditioned

$$\hat{\epsilon} = \epsilon_\theta(x_t, t, \emptyset) + w \cdot (\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \emptyset))$$

Classifier-Free Guidance

Methods

Parallel Denoising

Algorithm 1 Parallel Diffusion for Visual Anagram

$x \leftarrow \textsc{GenerateRandomNoise}()$
for $t = T-1, T-2, \ldots, 0$ **do**
    $\hat{\epsilon}_1 \leftarrow \textsc{UNet}(x, t, \text{“prompt 1”})$
    $x^{\text{rot}} \leftarrow \textsc{Transform}(x)$
    $\hat{\epsilon}_2 \leftarrow \textsc{UNet}(x^{\text{rot}}, t, \text{“prompt 2”})$
    $\hat{\epsilon}_2^{\text{align}} \leftarrow \textsc{Inverse}(\hat{\epsilon}_2)$
    $\hat{\epsilon} \leftarrow (\hat{\epsilon}_1 + \hat{\epsilon}_2^{\text{align}})/2$
    $x \leftarrow \textsc{RemoveNoise}(x, \hat{\epsilon}, t)$
end for
return x

$$\tilde{\epsilon}_t = \frac{1}{N} \sum_i v_i^{-1} \left(\epsilon_\theta(v_i(\mathbf{x}_t), y_i, t) \right).$$

Visual Anagrams

Core Idea: Orthogonal geometric transformations

$$\tilde{\epsilon} = \sum f_i(\epsilon_i).$$

Factorized Diffusion

Core Idea: Image is the sum of distinct linear components

Methods

Negative Prompting:

$$\hat{\epsilon} = \epsilon_\theta(x_t, t, \emptyset) + w \cdot (\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \emptyset))$$

$$\downarrow$$

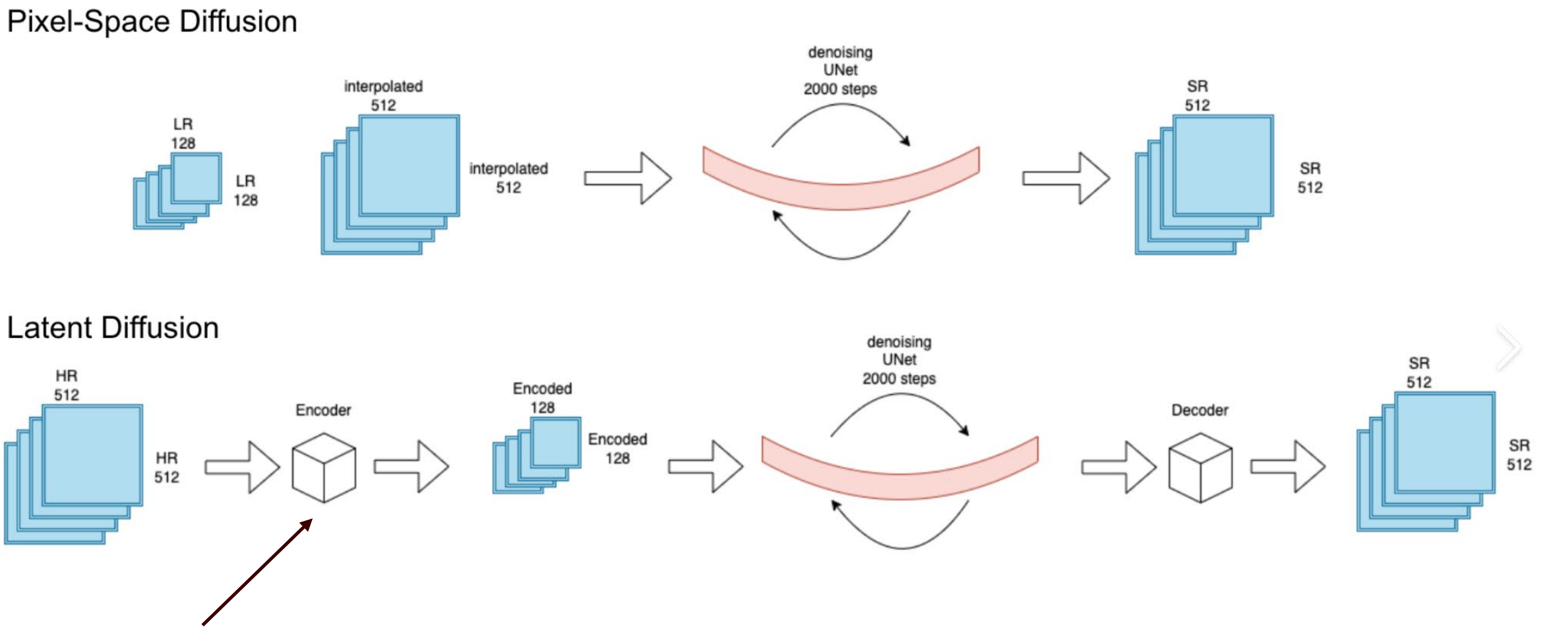
$$\hat{\epsilon} = \epsilon_\theta(x_t, c_{neg}) + w \cdot (\epsilon_\theta(x_t, c_{pos}) - \epsilon_\theta(x_t, c_{neg}))$$

Table 2. **Ablations.** We ablate negative prompting, reduction methods, and guidance scales on our dataset.

Ablation	$\mathcal{A}\uparrow$	$\mathcal{A}_{0.9}\uparrow$	$\mathcal{A}_{0.95}\uparrow$	$\mathcal{C}\uparrow$	$\mathcal{C}_{0.9}\uparrow$	$\mathcal{C}_{0.95}\uparrow$
Negative Prompting	0.24	0.27	0.276	**0.576**	**0.659**	**0.683**
No Negative Prompting	**0.255**	**0.285**	**0.295**	0.567	0.643	0.679
Alternating Reduction	0.252	**0.286**	0.292	0.560	0.639	0.664
Mean Reduction	**0.255**	0.285	**0.295**	**0.567**	**0.643**	**0.679**
$\gamma = 3.0$	0.239	0.271	0.285	0.537	0.610	0.629
$\gamma = 7.0$	0.255	0.285	0.295	0.567	0.643	0.679
$\gamma = 10.0$	**0.259**	**0.290**	**0.297**	**0.576**	**0.664**	**0.702**

Methods

Why Latent Diffusion Fail?



Variational Autoencoder: A non-linear neural network to encode for latent space

$$D(T(z)) \neq T(D(z))$$

Results

Visual Anagrams

a drawing
of a penguin

a drawing
of a giraffe

an oil painting of
Abraham Lincoln

an oil painting of
a woman staring
out a window

Factorized Diffusion

a lithograph of houseplants

a photo of new york city

Limitations

- Heavy. Details matter in illusion picture generation, but pixel diffusion models are large and heavy.

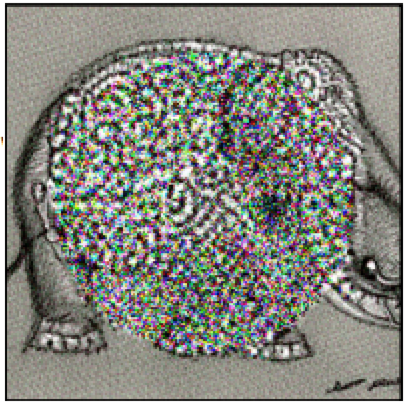
For Factorized Diffusion:

- High quality results are rare.
- Sensitive to parameters: One prompt could dominate the generated image easily
- Color and Contrast Degradation: Because the method sums extracted noise signals from different frequency bands or color channels, the final mathematical result often averages out the pixel intensities.

Limitations

For Visual Anagrams:

- Independent Synthesis: The model synthesizing prompts separately, without combining elements of the two to form an illusion.
- Correlated Noise. Must insure the transformation does not introduce correlations in the noise
- Not for all illusions, such as color constancy illusions, homographies, stretches, and more generally nonvolume-preserving deformations.



Failure: Correlated Noise

View: Rotate Inner Circle 45 Degrees (Bilinear)

`a sketch of an elephant`

`a sketch of a mouse`

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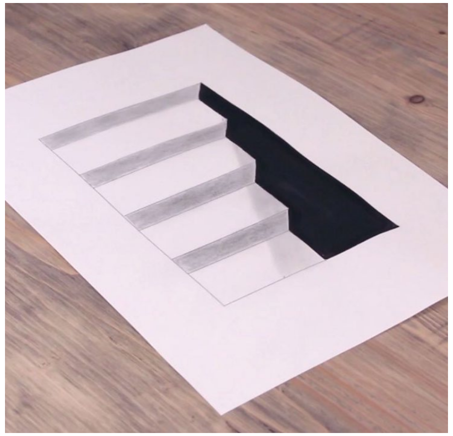
Future Works



- Other illusion generations

- Train a VAE that keeps structural consistency

$$D(T(z)) = T(D(z))$$

- Lift 2D illusion to 3D illusion generation.

Thank You!

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A FORCE FOR GOOD